Topic 13 | Evidence Report

Error Handling, Retries & Recovery Strategies - n8n

Assessment evidence based on screenshots supplied during workflow implementation and testing on 23 October 2026.

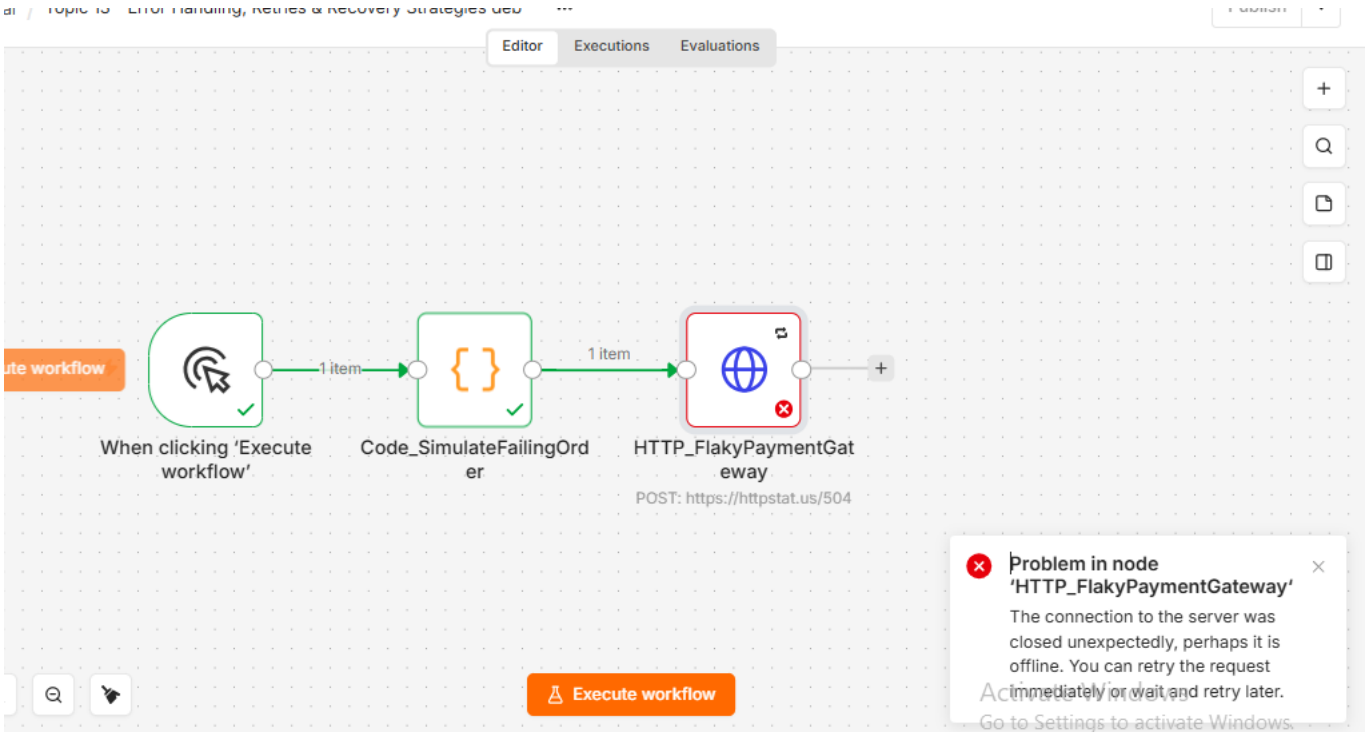
Scenario	Observed result
Production gateway failure	Connection reset / HTTP node failed
Error classification	ECONNRESET -> TRANSIENT
Transient path	RETRY_SCHEDULED; PENDING_RETRY
Permanent test	PERMANENT -> ESCALATED_TO_ENGINEERING
Audit outputs	Both paths generated RECOVERY_AUDIT records

Scope note. The DLQ and replay scheduling are represented by Code node output records; the supplied evidence does not establish persistent DLQ storage, an actual delayed replay, or delivery of an engineering notification. The permanent scenario uses simulated test data.

The following pages preserve the submitted screenshots and explain each result.

Evidence 01 | Primary workflow and failed gateway

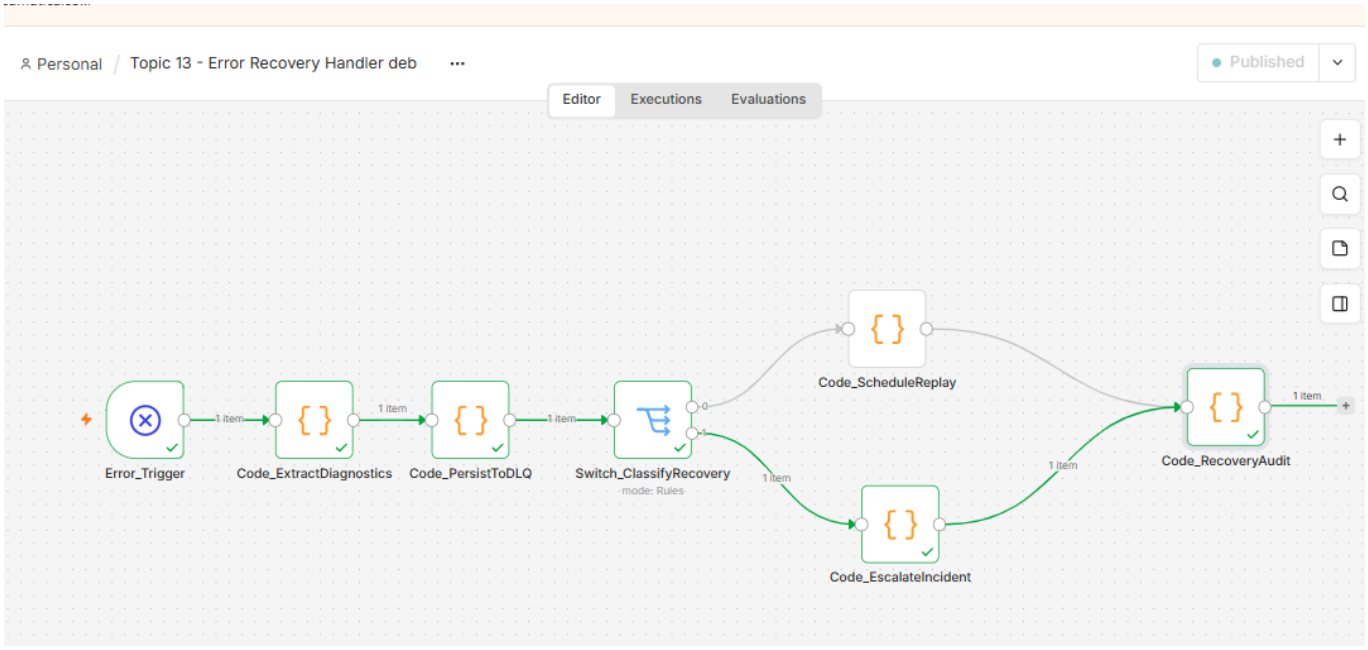
The main workflow generates a sample payment and calls HTTP_FlakyPaymentGateway. The HTTP request fails with a connection error, providing the failure condition for recovery testing.



Source: user-provided n8n screenshot (20261023-103528).

Evidence 02 | Error handler workflow - initial routing

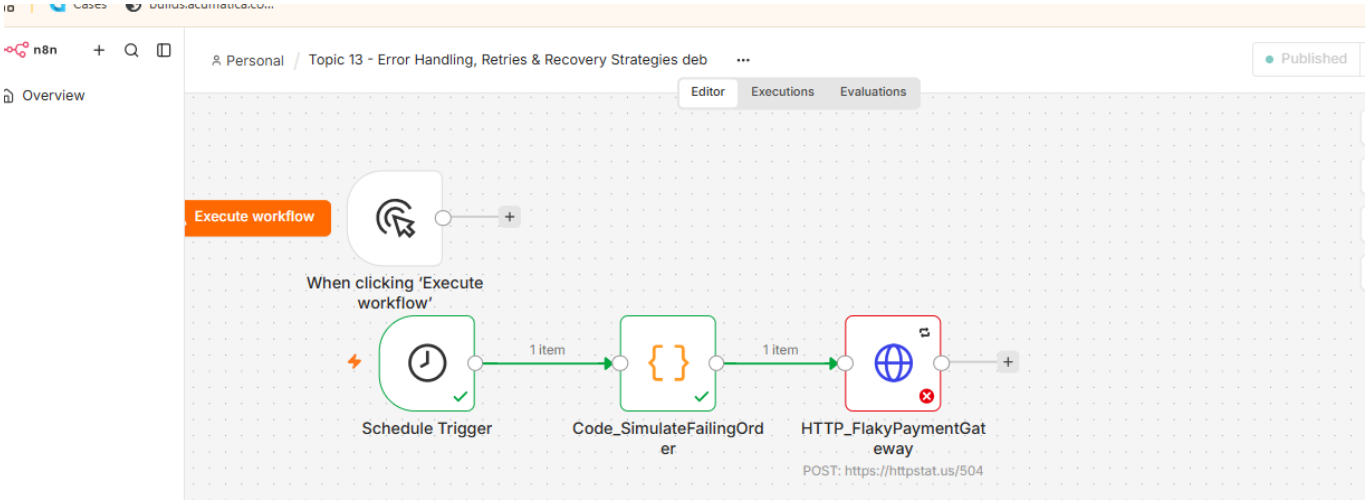
The error handler contains Error Trigger, diagnostics, DLQ record creation, classification, recovery actions, and final audit. This screenshot shows the earlier permanent route before the real gateway error was tested.



Source: user-provided n8n screenshot (20261023-103458).

Evidence 03 | Main workflow - scheduled failure

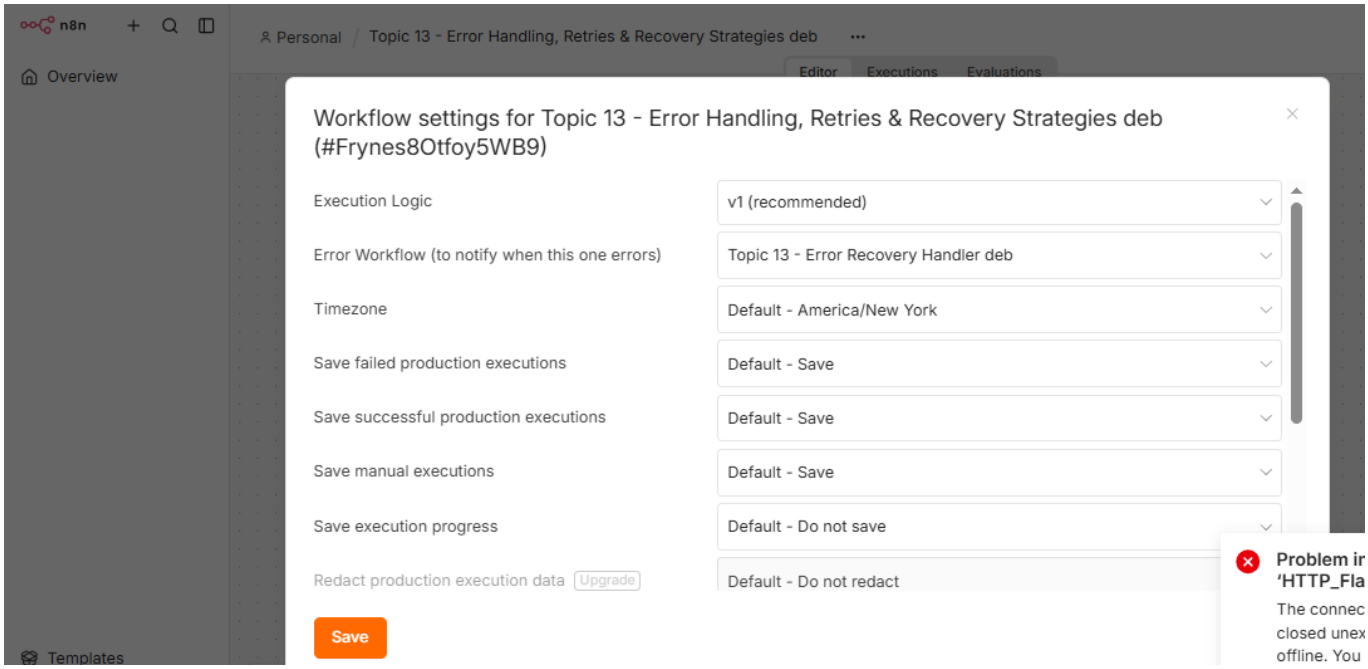
The temporary Schedule Trigger runs the primary workflow automatically. The sample order node succeeds and the gateway request fails, allowing a production execution to invoke the configured error workflow.



Source: user-provided n8n screenshot (20261023-104000).

Evidence 04 | Recovery handler configuration

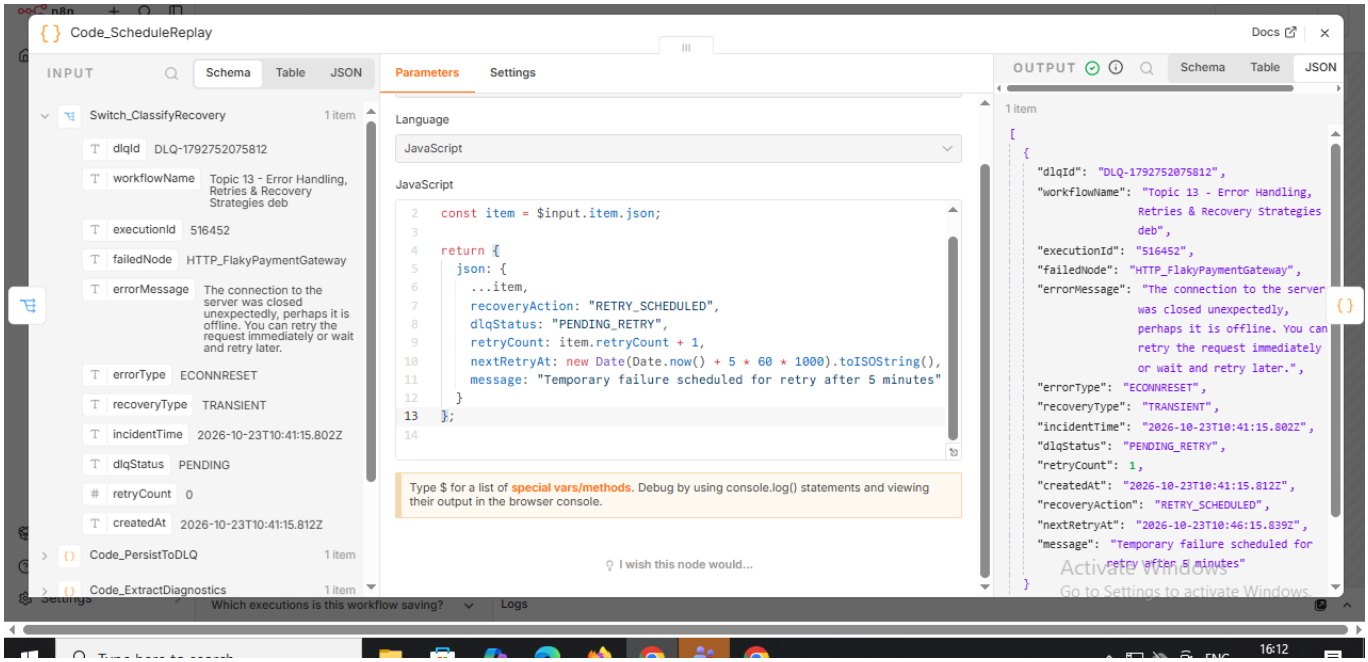
The main workflow settings select Topic 13 - Error Recovery Handler deb as its Error Workflow, linking production failures to the dedicated handler.



Source: user-provided n8n screenshot (20261023-103749).

Evidence 05 | Transient branch - replay preparation

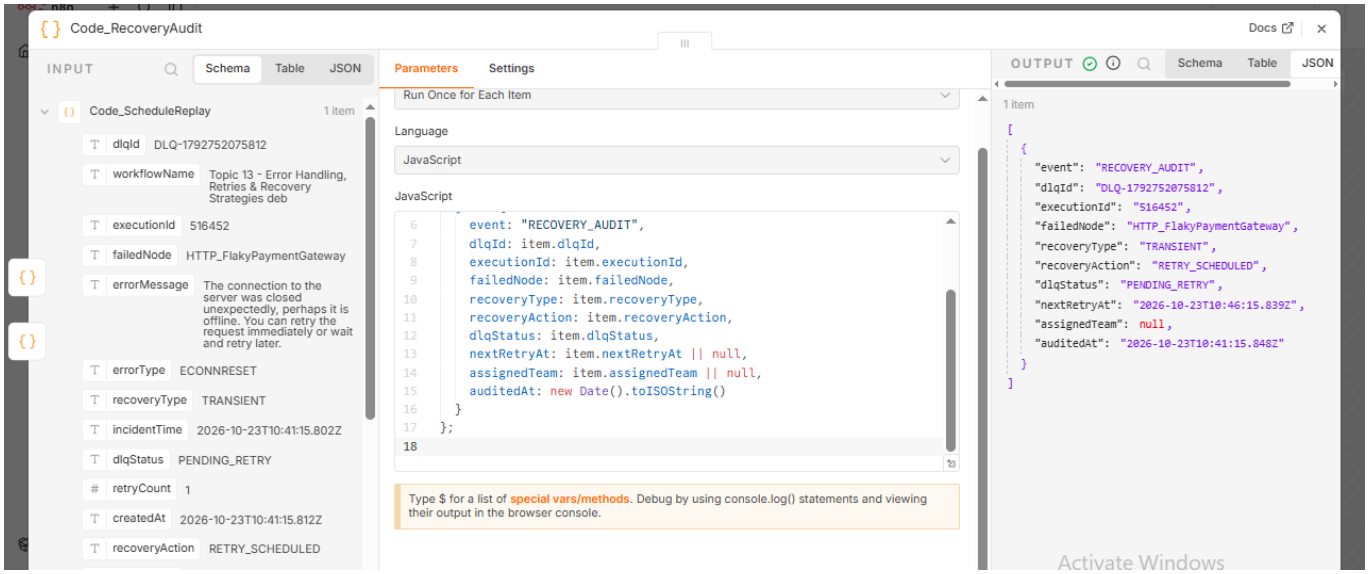
The actual ECONNRESET error is classified as TRANSIENT. Code_ScheduleReplay produces RETRY_SCHEDULED, PENDING_RETRY, retryCount 1, and a nextRetryAt timestamp five minutes later.



Source: user-provided n8n screenshot (20261023-104219).

Evidence 06 | Transient branch - recovery audit

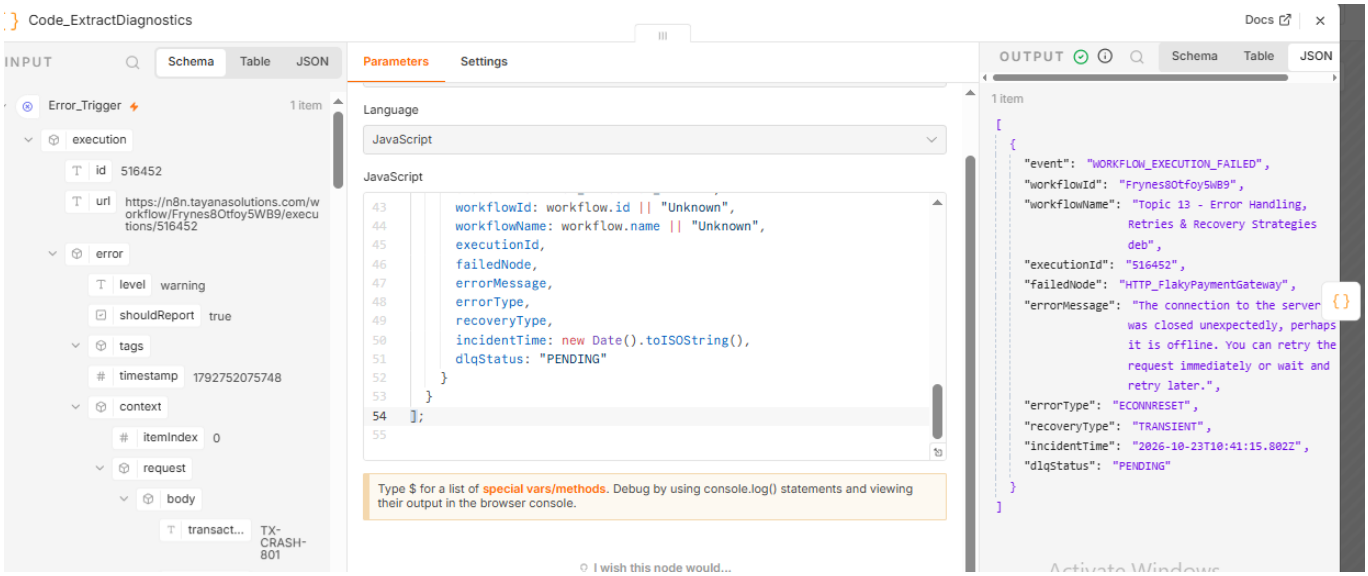
The audit records execution 516452, failed node HTTP_FlakyPaymentGateway, TRANSIENT classification, RETRY_SCHEDULED action, and PENDING_RETRY status.



Source: user-provided n8n screenshot (20261023-104254).

Evidence 07 | Diagnostics from actual failure

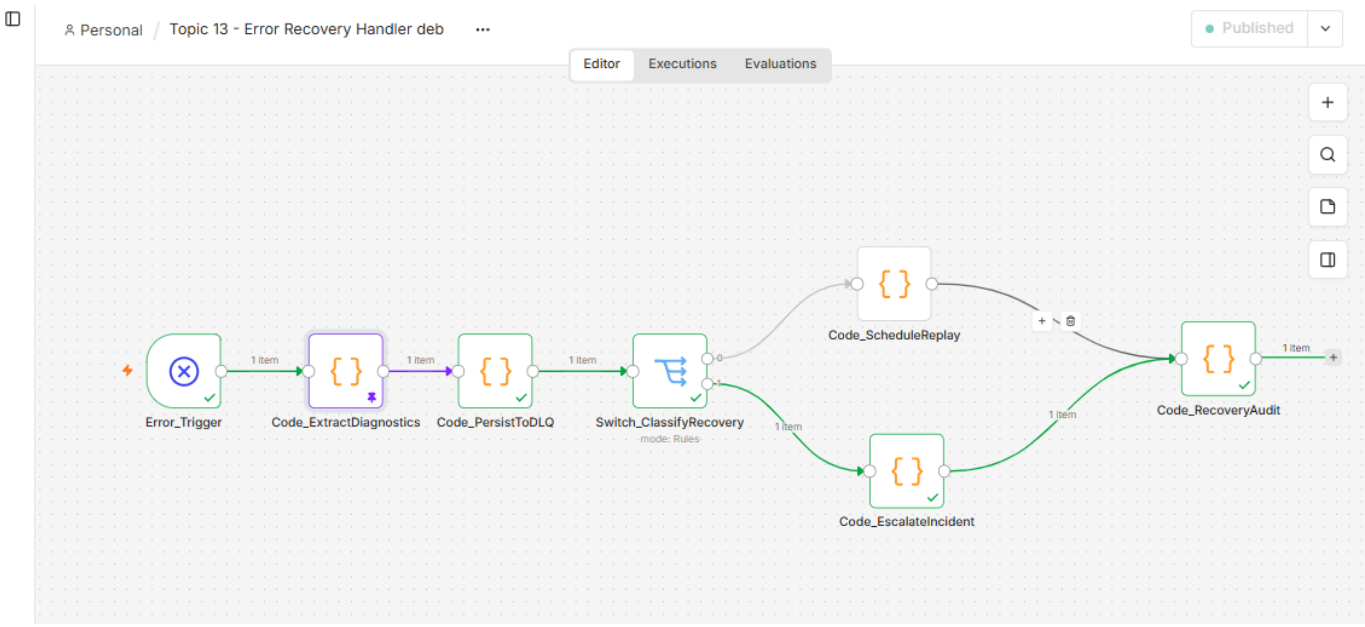
Code_ExtractDiagnostics receives a real Error Trigger payload and extracts workflow details, failed node, ECONNRESET, and TRANSIENT classification instead of placeholder sample data.



Source: user-provided n8n screenshot (20261023-104338).

Evidence 08 | Permanent branch - workflow routing

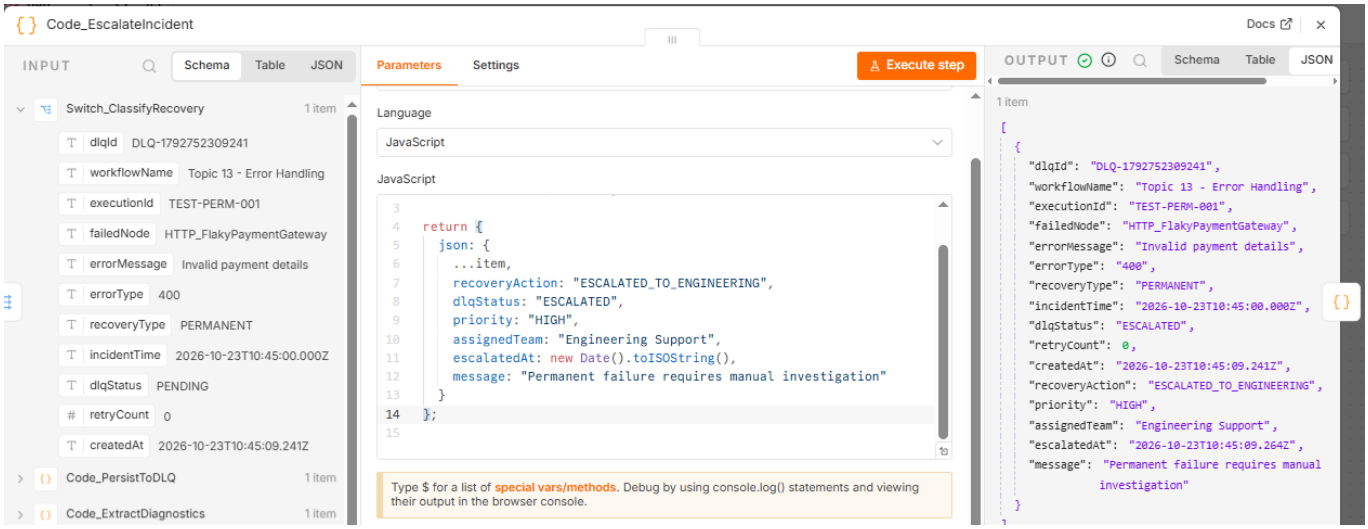
Pinned test data simulates invalid payment details. The Switch routes the test to Code_EscalateIncident rather than Code_ScheduleReplay, and the final audit node executes.



Source: user-provided n8n screenshot (20261023-104527).

Evidence 09 | Permanent branch - escalation output

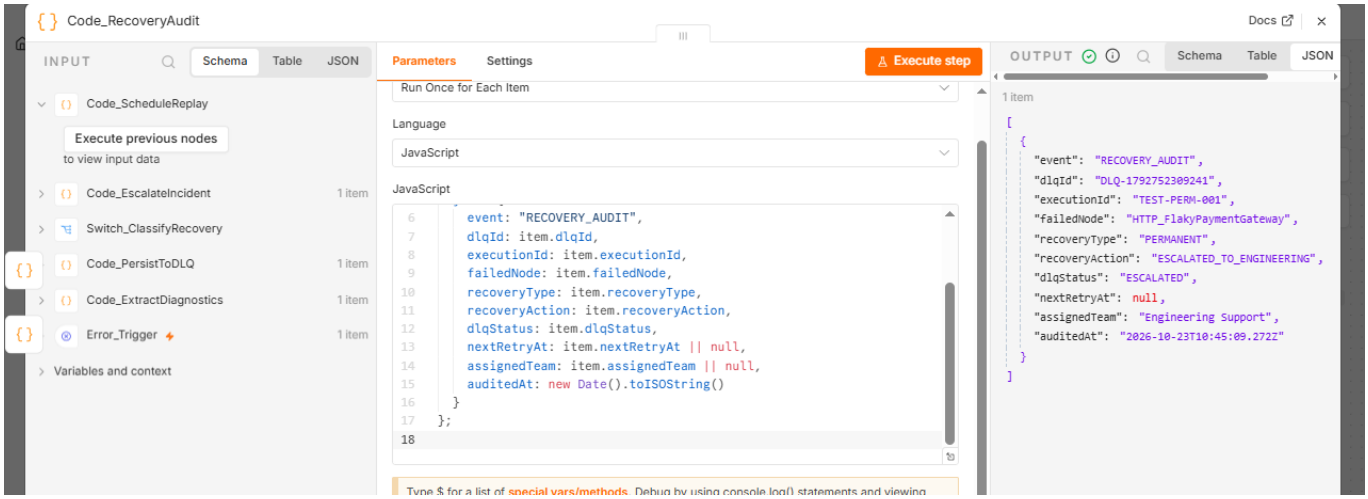
Code_EscalateIncident records test execution TEST-PERM-001, recoveryType PERMANENT, ESCALATED_TO_ENGINEERING, priority HIGH, and Engineering Support assignment.



Source: user-provided n8n screenshot (20261023-104605).

Evidence 10 | Permanent branch - final audit

Code_RecoveryAudit records TEST-PERM-001, PERMANENT classification, ESCALATED_TO_ENGINEERING action, ESCALATED status, and Engineering Support assignment.



Source: user-provided n8n screenshot (20261023-104622).